Screening the Slochteren Aquifer for Urban District Heating and Cooling in Utrecht, Netherlands: A Data-Driven Doublet Assessment

Kadi Sadaraka

October 2026

Abstract

This paper presents a geoscientific and technoeconomic screening workflow for evaluating geothermal development opportunities in the Utrecht region. The study integrates well-log interpretation, reservoir characterization, spatial analysis, and scenario-based economic optimization to identify promising geothermal targets and quantify their commercial viability. The repository associated with this project contains raw and processed geological data, exploratory notebooks, and a Python-based levelized cost of energy (LCOE) model. The work combines petrophysical diagnostics from LAS well logs, ThermoGIS-derived subsurface property databases, and multi-criteria decision logic for the identification of candidate geothermal scenarios.

A single stimulated doublet at BLT-01 can plausibly supply the 10 MWth heating target for Utrecht, but with roughly even odds and at a cost close to the gas alternative. The preferred design is one stimulated doublet at BLT-01 with a 60 bar drawdown and 211 m3/h flow, giving 12.2 MWth in the deterministic case, with surface electric chillers for the 5 MWth cooling load. A Monte Carlo run over ThermoGIS transmissivity gives a 56% probability that the wells alone reach 10 MWth, a median of 12.7 MWth and a low case (P90) of 1.7 MWth. Whole-system capital cost is 23.3 M€ and the levelised cost is 69 €/MWh against 68 €/MWh for a gas and chiller baseline. Transmissivity uncertainty at BLT-01 is the dominant risk, and a pilot test of that well is the logical next step.

Keywords: geothermal energy, Slochteren Formation, Rotliegend, doublet, district heating, district cooling, ThermoGIS, Utrecht, techno-economic screening

1. Introduction

Space heating and hot water account for a large share of final energy demand in northern European cities, and in the Netherlands most of it is still met with natural gas. Replacing it needs low-carbon, local sources that connect to district heating networks. The Netherlands has one of Europe's most active geothermal sectors, supported by a national subsurface information system.

Utrecht is a fast-growing, densely built city that wants to heat and cool its building stock without fossil fuels. Its deep subsurface contains the Slochteren Formation, a Lower Permian Rotliegend sandstone known from decades of hydrocarbon exploration and recognised as a geothermal target. Whether it can deliver the power a district network needs, at viable cost, depends on depth, thickness, porosity and permeability, which are measured directly at only a few wells.

This paper addresses a screening question posed by the SPE Africa Datathon 2026: can the Slochteren aquifer (ROSL_ROSLU) deliver at least 10 MWth for district heating and at least 5 MWth for district cooling using commercially viable doublet configurations, and which enhancement scenario and locations meet these targets economically?

The contributions are:

1. A reproducible four-notebook workflow from raw well logs and ThermoGIS grids to a costed system design.
2. Recovery and quality control of the well dataset, including conversion of along-hole depths to true vertical depth and a leave-one-well-out test of porosity prediction.
3. A multi-criteria ranking of four wells by reservoir quality, with sensitivity tests on weights and geothermal gradient.
4. A Monte Carlo comparison of four well-intervention strategies under ThermoGIS transmissivity uncertainty, and a system-level cost comparison of 32 design scenarios.

2. Background

2.1 Geological setting

The Slochteren Formation is the main reservoir of the Upper Rotliegend Group in the Netherlands. It consists mainly of aeolian and fluvial sandstones deposited in a Permian desert basin, and it hosts the Groningen gas field and many smaller fields. It therefore has a far denser legacy of seismic and well data than most geothermal targets. In ThermoGIS nomenclature, the unit assessed here is ROSL_ROSLU, the Slochteren Formation together with its Upper Slochteren Member.

Reservoir quality is controlled by depositional facies and burial diagenesis. Porosity and permeability change over short distances and generally decline with depth. Thermal power depends on both the temperature of the produced fluid, which rises with depth, and the flow rate the reservoir can sustain, which depends on permeability and thickness. Screening has to weigh these competing trends.

2.2 Doublet performance

A doublet has a production well that brings hot water to the surface and an injection well that returns the cooled water to the aquifer. Thermal power is

$$P_{th} = \dot{m}\, c_p (T_p - T_{inj})$$

where ṁ is the mass flow rate, c_p the specific heat of water, T_p the production temperature and T_inj the reinjection temperature.

Flow rate is limited by transmissivity (permeability times thickness), by the pressure drawdown allowed, and by pump capacity. Flow is modelled with radial Darcy flow between producer and injector, and long-term thermal and pressure behaviour is checked with a cold-front radius and the Theis solution (Section 4.6).

2.3 Heating, cooling and the datathon targets

The targets are at least 10 MWth for heating and at least 5 MWth for cooling. This study treats them differently. Heat is supplied from the aquifer. Cooling is supplied by a separate surface

plant, and heating and cooling are assumed not to occur at the same time (Section 4.8). Section 6.3 discusses the consequence of that choice.

2.4 ThermoGIS

ThermoGIS is the Dutch geothermal information system developed by TNO. It provides gridded aquifer properties and P90, P50 and P10 estimates that express subsurface uncertainty. This study uses the ThermoGIS grids (version 2.5.1) for formation coverage and depth mapping, and the P90, P50 and P10 values at each well for permeability and transmissivity.

3. Data

3.1 Datasets

The analysis uses four wells supplied in the datathon data pack, together with public boundary and grid data.

- **Well logs and tables.** LAS logs for BLT-01, JUT-01, EVD-01 and PKP-01, a table of target lithology samples (3,455 rows), well path surveys, and ThermoGIS property sheets per well.
- **Administrative boundaries.** Municipality (Utrecht, GM0344) and province (PV26) polygons from the Dutch PDOK service, reprojected from EPSG:4258 to RD New (EPSG:28992).
- **ThermoGIS grids.** NetCDF grids from the 6_Permian folder of ThermoGIS release 2.5.1, 132 files in all, of which the 11 BaseCase ROSL_ROSLU grids were exported.

Table 1 Wells used in the study. Coordinates and permeability are from the ThermoGIS sheets

Well	Easting (RD New, m)	Northing (RD New, m)	Samples used	Porosity source	ThermoGIS permeability P50 (mD)
BLT-01	141,578	456,882	1,689	Measured	82
JUT-01	134,098	451,726	836	Predicted	40
EVD-01	136,997	441,189	780	Predicted	6

PKP-01	118,503	453,403	730	Measured	1

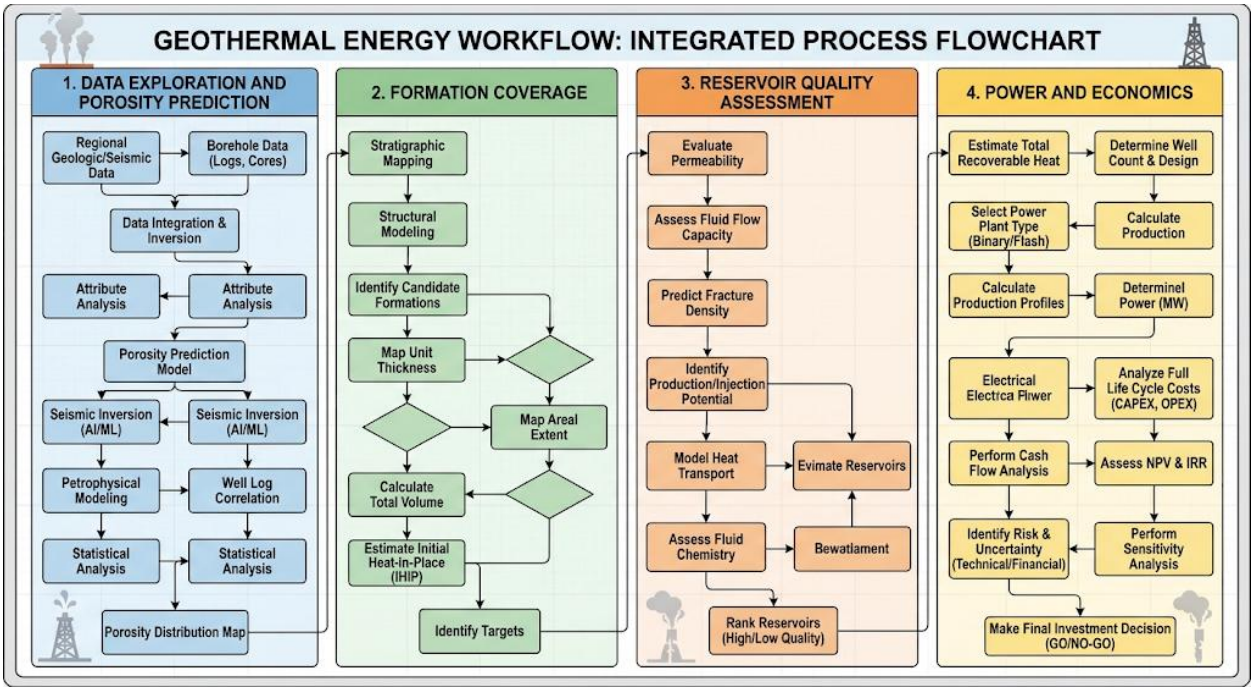
3.2 Data quality issues and fixes

Several problems in the supplied data had to be resolved before analysis.

- **Depth reference.** Every sample in the target table was flagged as an along-hole depth in a deviated well, and the true vertical depth column was empty. Formation tops and bases were converted by interpolating each well path survey, and sample depths were recovered by matching gamma ray values to the LAS logs. All 1,689, 780 and 730 rows for BLT-01, EVD-01 and PKP-01 matched.
- **JUT-01 window.** The 256 JUT-01 rows in the table fell in the wrong depth window. They were replaced with 836 rows rebuilt from the LAS logs inside the correct interval, giving 4,035 samples in total.
- **Missing porosity.** Porosity was measured for only BLT-01 and PKP-01 (2,419 samples). EVD-01 and JUT-01 had none, and were filled by prediction (Section 4.2).
- **Missing density.** Bulk density for JUT-01 was missing in the original table and was restored from the RHOB log.

4. Methodology

The workflow runs as four notebooks: data exploration and porosity prediction, formation coverage, reservoir quality assessment, and power and economics. Processed tables and results are saved between stages so each can be rerun alone. [Figure 1: workflow diagram from raw data to costed design.]



4.1 Well-log preprocessing

LAS logs were read with lasio, converted to metric along-hole depth, sorted, and the sonic curve was standardised to DTC. Along-hole depths were converted to true vertical depth (TVD) by linear interpolation of each well path survey. Sample depths in the target table were recovered by matching each sample's gamma ray value to the LAS curve inside the formation window, as described in Section 3.2.

4.2 Porosity estimation

Two wells (BLT-01 and PKP-01) have measured porosity. For the other two, porosity was predicted from bulk density and sonic transit time. Two physics baselines were computed first: density porosity with matrix densities of 2.68 g/cm^3 (BLT-01) and 2.71 g/cm^3 (PKP-01) and fluid density 1.07 g/cm^3, and Wyllie time-average sonic porosity with matrix transit times of 50 to 55 µs/ft and fluid transit time 190 µs/ft.

Six candidate predictors were then compared by leave-one-well-out cross-validation: a per-well median, the density and sonic baselines, linear regression, a Random Forest (100 trees, depth 4) and gradient boosting (XGBoost, 100 trees, depth 4, learning rate 0.01). Each fold trains on one well and tests on the other. Features were standardised. The best method was linear regression and it was refitted on both wells and applied to EVD-01 and JUT-01.

4.3 Formation coverage

The ThermoGIS Permian grids were searched for Slochteren units overlapping a bounding box around Utrecht province (easting 110 to 180 km, northing 420 to 490 km in RD New). Presence was tested by counting valid grid cells with positive values inside the box, using xarray and geopandas.

4.4 Temperature and permeability

Reservoir temperature follows a linear gradient from a 10 °C surface temperature at 31 °C/km, the regional value. The BLT-01 bottom-hole temperature header (167 °F at 2,123.8 m) gives a consistent gradient of 30.6 °C/km. A Harrison bottom-hole temperature correction was added to the modelled temperature, and the mean per well was taken as reservoir temperature. Permeability is the ThermoGIS P50 value at each well (Table 1).

4.5 Reservoir quality ranking

Wells were scored on six criteria: porosity, thickness, temperature and permeability (log-transformed) as benefits, and gamma ray and depth as costs. Criteria weights come from the entropy weight method, which gives more weight to criteria that differ more between wells. Scores were then computed three ways: simple additive weighting (SAW), TOPSIS, and the first principal component of a PCA on all samples. K-means, hierarchical and Bayesian Gaussian mixture clustering were used as an exploratory check. Classes (poor, marginal, good, excellent) were assigned by score quartile.

Three sensitivity tests checked the SAW ranking: 500 random perturbations of the weights by ±20%, replacement of entropy weights by equal and CRITIC weights, and a ±10% change in geothermal gradient.

4.6 Doublet performance and interference

Flow rate for a single well pair uses radial Darcy flow,

$$Q = \frac{2\pi k h \Delta P}{\mu[\ln(r_e/r_w)+S]}, \qquad \mu = 2.414 \times 10^{-5} \cdot 10^{247.8/(T+133.15)}$$

with drainage radius r_e of 1,000 m, wellbore radius r_w of 4 inches, skin S of −1 and drawdown ΔP of 18 bar in the base case, and reinjection temperature of 40 °C. The system design (Section

4.8) replaces this with a producer and injector resistance in series at 1,500 m spacing, capped by a pump limit of 250 m^3/h.

Long-term behaviour was checked at 150 m3/h over 30 years. The cold-front radius is

$$R_{th} = \sqrt{\frac{Qtc_w}{\pi H c_{aq}}}$$

and the breakthrough time uses a Gringarten-type expression, with volumetric heat capacities of 4.18 and 2.3 MJ/m^3/K for water and saturated rock. Pressure interference between the BLT-01 and JUT-01 doublets was computed with the Theis solution using a specific storage of 10^{-6} per metre.

4.7 Intervention strategies and Monte Carlo

Four strategies were defined per well. **None** uses skin −1 and reinjection at 30 °C. **Stimulation** (named thermogis_stim in the code) lowers skin to −4. **Heat pump** lowers reinjection temperature to 20 °C with a coefficient of performance of 5. **Stimulation plus heat pump** combines both.

Transmissivity was sampled 10,000 times from a lognormal distribution fitted to the ThermoGIS P90, P50 and P10 values, with log-standard deviations of 1.53 for BLT-01 and 0.61 for JUT-01. Every subset of drilled wells and every strategy combination was evaluated, and ranked by the probability of reaching 10 MWth.

4.8 System design, cooling and economics

The system design compared 32 scenarios: drawdown of 18, 30, 45 or 60 bar, one or two doublets, with and without stimulation, with and without a heat pump. Each was evaluated with 300 Monte Carlo draws. Scenarios were kept if the wells alone reached 10 MWth in at least half of the draws, then ranked by capital cost.

Heat demand is 10 MWth at peak with a load factor of 0.43 (37.7 GWh per year). Cooling demand is 5 MWth at peak with a load factor of 0.2 (8.8 GWh per year). Cooling is provided by electric chillers with a coefficient of performance of 4.5 and a 6 K loop temperature difference. The code also defines an aquifer thermal energy storage option (coefficient of performance 20),

which was not selected. Cost and financing assumptions are given in Section 4.9. The reference alternative is a gas boiler plus electric chillers.

4.9 Cost model

Costs come from a Python translation of the TNO direct-heat levelised cost spreadsheet (van Wees, ten Kulve, Straathof and Pluymaekers, version 2012.1), whose cash-flow method follows ECN models. It was extended to price heat and cooling as one system. All capital is spent in year 0 and operating costs are inflated at 2% a year. The levelised cost is the flat price in €/MWh that makes the equity investor's net present value, discounted at the required equity return, equal to the equity outlay. It covers heat and cooling together.

Table 2 Cost and financing assumptions

Item	Assumption
Drilling	375,000 + 1,150 d + 0.3 d² € per well (d = depth in m), ThermoGIS 2019 calibration, escalated by 1.25 to 2026 prices
Stimulation	0.25 M€ per well
Pumps	0.3 M€ per well, replaced every 5 years
Surface plant	3 M€ per doublet plus 300 €/kW installed thermal power, escalated by 1.25
Contingency	15% on wells, pumps and surface plant
Geothermal operating cost	60 €/kW per year plus 1.9 €/MWh of heat
Heat pump	200 €/kW capital, 20 €/kW per year operating
Backup gas boiler	0.1 M€/MW; gas at 60 €/MWh of heat
Electric chiller	0.3 M€/MW, no escalation applied
Electricity	80 €/MWh
Aquifer thermal energy storage (not selected)	300 €/kW installed; operating cost 3% of capital per year
Financing	20% equity, 80% loan over 15 years at 5%; equity return 7%; tax 25%; 15-year depreciation; 30-year life

Payback is computed from the same cash flow with a selling price of 68 €/MWh, the levelised cost of the gas and electric chiller baseline, which is priced with the same engine.

5. Results

5.1 Porosity prediction

Linear regression on bulk density and sonic transit time was the best predictor on both held-out wells. For BLT-01, it gave a mean absolute error of 1.19 porosity points and R^2 of 0.72. For PKP-01, the error was similar (1.16 points) but R^2 was −0.08, so the model captured the well's average level and almost none of its variation. PKP-01 porosity varies little (standard deviation about 1.95 points), and its density-to-porosity relationship is weaker (correlation −0.47 vs. −0.95 at BLT-01). Random Forest and gradient boosting did worse on both wells, consistent with training on a single well and limited extrapolation.

Table 3 Leave-one-well-out cross-validation of porosity prediction

Method	**MAE BLT-01**	**MAE PKP-01**	**RMSE BLT-01**	**RMSE PKP-01**	**R^2 BLT-01**	**R^2 PKP-01**
Linear regression	1.19	1.16	1.57	2.02	0.72	−0.08
Density porosity	1.92	1.94	2.20	2.75	0.45	−0.99
Sonic porosity	1.45	3.11	1.85	3.38	0.61	−2.01
Random Forest	3.23	1.96	3.96	2.64	−0.79	−0.84
Gradient boosting	2.22	4.04	2.84	4.40	0.08	−4.11
Per-well median	7.23	7.37	7.81	7.57	−5.94	−14.10

The predicted porosity carries an uncertainty of 2.02 points, the larger of the two linear regression RMSE values. Mean porosity is 16.4% at BLT-01, 12.6% at JUT-01 (predicted), 9.1% at EVD-01 (predicted) and 9.0% at PKP-01.

5.2 Formation coverage

The Lower Slochteren Member (ROSLL) does not reach the Utrecht area, since its grid lies far to the north. The combined Slochteren Formation and Upper Slochteren Member (ROSL_ROSLU)

is present, with 4,405 valid grid cells inside the Utrecht bounding box. All later analysis therefore uses ROSL_ROSLU.

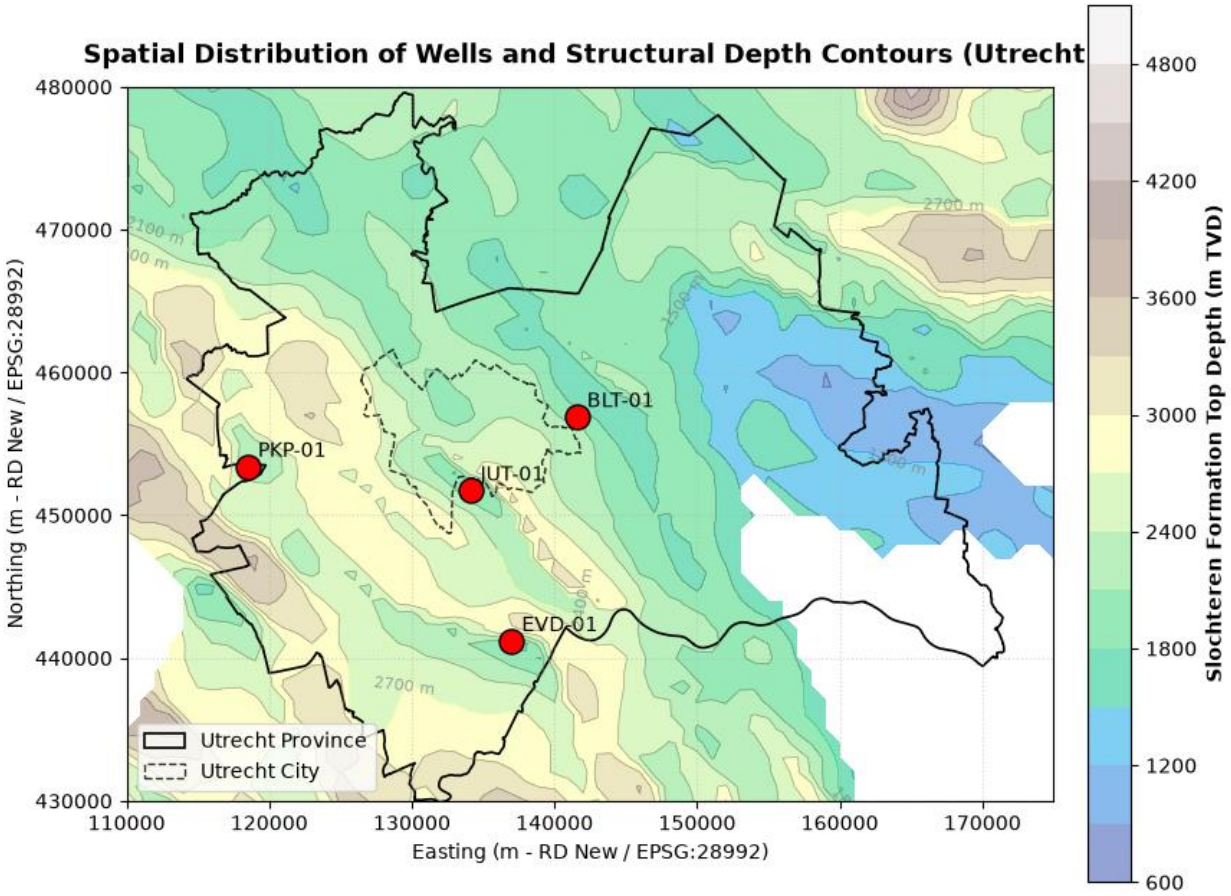


Figure 1 Formation Top Depth Map with wells

5.3 Reservoir quality

BLT-01 ranks first and JUT-01 second under SAW and TOPSIS, EVD-01 is third and PKP-01 last. PCA gives the same grouping but puts JUT-01 ahead of BLT-01. Two principal components explain 92.4% of the variance (64.5% and 27.9%). Entropy weights are largest for porosity (0.27), then temperature (0.18), thickness (0.17), permeability and gamma ray (0.13 each) and depth (0.12).

Table 4 Reservoir quality scores by well

Well	Mean porosity (%)	Mean gamma ray (API)	Permeability P50 (mD)	SAW	TOPSIS	PCA	SAW class
BLT-01	16.4	49.0	82	0.755	0.775	0.73	Excellent
JUT-01	12.6	24.3	40	0.657	0.703	2.03	Good
EVD-01	9.1	28.5	6	0.340	0.389	−0.15	Marginal
PKP-01	9.0	65.0	1	0.182	0.137	−3.86	Poor

The ranking did not change in any sensitivity test. All 500 weight perturbations kept the order BLT-01, JUT-01, EVD-01, PKP-01. Equal and CRITIC weights gave the same order (rank correlation 1.0), as did gradient changes of ±10%. Hierarchical clustering separated PKP-01 from the rest, and K-means with two clusters grouped BLT-01 with JUT-01 (silhouette 0.28). BLT-01 and JUT-01 were selected for power assessment.

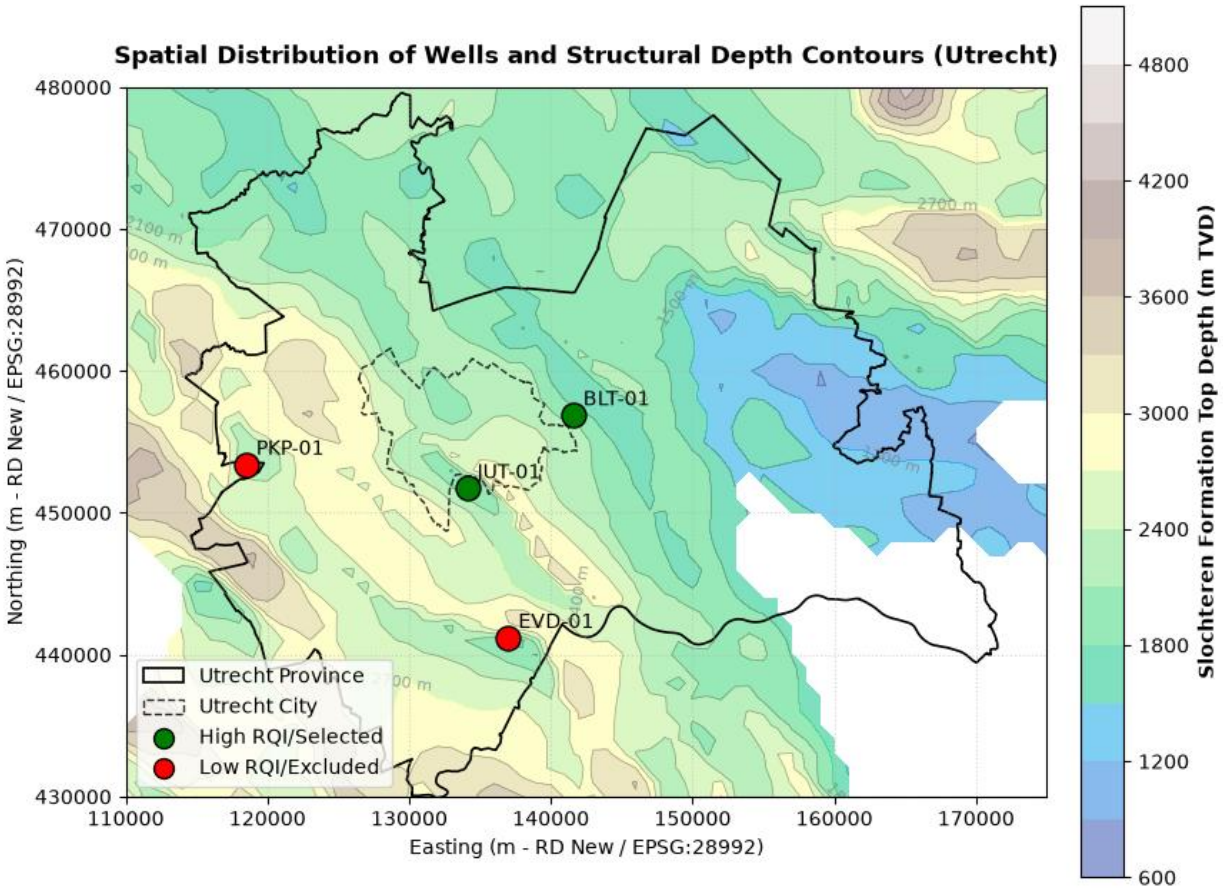


Figure 2 Selected wells with high reservoir quality

5.4 Base-case power and interference

With no intervention, BLT-01 delivers 6.25 MWth and JUT-01 2.21 MWth. BLT-01 therefore exceeds the 5 MWth threshold but falls 3.75 MWth short of the heating target, and JUT-01 meets neither.

The two doublets are 9,084 m apart, with 7,895 m between the nearest wells. At 150 m3/h over 30 years the cold front advances about 430 m at both wells (432 m at BLT-01, 426 m at JUT-01), well short of the 750 m half-spacing, and the thermal breakthrough time is 121 to 124 years. The Theis calculation gives hydraulic interference of about 0.45 to 0.49 bar. The doublets do not interfere thermally and interfere hydraulically only slightly, so they can be treated as independent.

5.5 Well interventions under uncertainty

The transmissivity range at BLT-01 is very wide (P90 1.3, P50 9.3 and P10 66.1 darcy-metres), against 2.2, 4.8 and 10.4 for JUT-01. Stimulation plus a heat pump on both wells gives the highest probability of reaching 10 MWth at 87%, with a median of 23.8 MWth and a low case (P90) of 9.1 MWth.

Table 5 Selected combinations from the 24 strategy grid, ranked by probability of reaching 10 MWth

BLT-01 strategy	JUT-01 strategy	P(≥10 MWth)	P50 (MWth)	P90 low case (MWth)
Stimulation + heat pump	Stimulation + heat pump	0.87	23.76	9.10
Stimulation	Stimulation + heat pump	0.82	17.74	7.83
Heat pump	Stimulation + heat pump	0.81	19.84	7.70
Stimulation + heat pump	Stimulation	0.80	22.02	6.89
Stimulation + heat pump	Not drilled	0.64	16.83	2.29
None	None	0.53	10.89	3.76

None	Not drilled	0.41	7.10	0.97
Not drilled	Stimulation + heat pump	0.27	6.86	3.16

BLT-01 carries most of the result. Downgrading its strategy from stimulation plus heat pump costs 5 to 6 percentage points of probability, and leaving it undrilled drops the best case to 27%.

5.6 System design scenarios

Twelve of the 32 scenarios met the heating target in at least half of the draws. The cheapest is a single stimulated doublet at 60 bar drawdown with no heat pump.

Table 6 Selected feasible scenarios sorted by capital cost (8 of 12 shown). The gas and electric chiller baseline costs 68 €/MWh

Drawdown (bar)	Doublets	Stimulation	Heat pump	Flow (m3/h)	Wells power (MWth)	Capex (M€)	LCOE (€/MWh)	P(≥10 MWth)
60	1	Yes	No	211	12.17	23.29	69.2	0.53
45	1	Yes	Yes	134	11.64	25.39	91.9	0.58
60	1	Yes	Yes	179	15.52	27.84	104.6	0.62
60	2	No	No	203	11.07	37.02	92.4	0.59
30	2	Yes	No	157	8.53	37.07	86.5	0.51
45	2	Yes	No	235	12.80	38.77	96.9	0.62
60	2	Yes	No	313	17.07	40.61	107.7	0.75
60	2	Yes	Yes	266	22.01	47.14	152.1	0.83

Levelised cost rises with every added element. A second doublet raises capital cost by about 14 to 24 M€ for a higher probability of success, and no scenario beats the baseline.

5.7 Selected design

The selected design is one stimulated doublet at BLT-01, 60 bar drawdown, no heat pump, with 5 MWth of electric chillers for cooling. It delivers 211 m3/h and 12.2 MWth, with no backup boiler needed in the deterministic case. Annual output is 37.7 GWh of heat and 8.8 GWh of cooling. The 30-year cold front reaches 512 m at this flow rate.

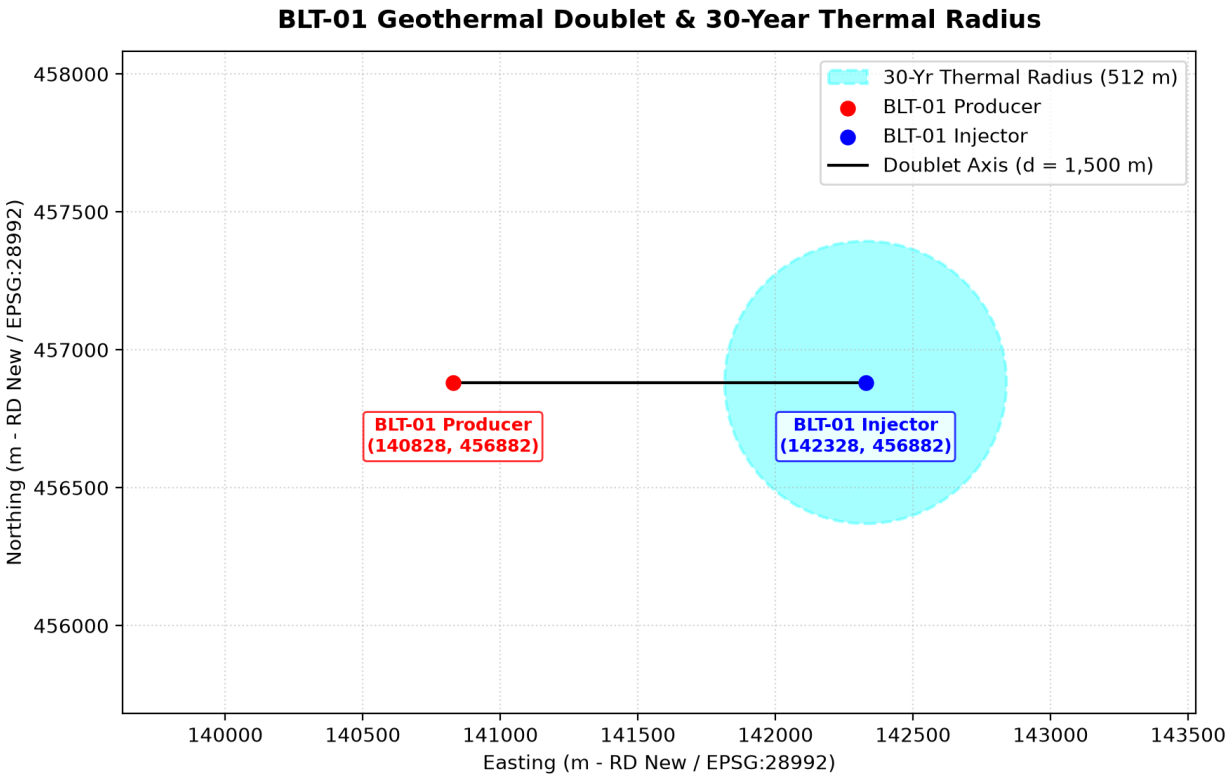


Figure 3 BLT-01 Doublet Site and Thermal Radius

Table 7 Cost of the selected design

Item	Value
Capital cost	23.3 M€ (geothermal 21.8, cooling 1.5)
Revenue needed	3.21 M€ per year for 46.4 GWh
Whole-system LCOE	69 €/MWh against a 68 €/MWh selling price
Equity payback	12.0 years simple, 16.7 years discounted

A 2,000-draw Monte Carlo using the BLT-01 transmissivity spread gives a 56% probability that the wells alone reach 10 MWth. The median is 12.7 MWth and the low case (P90) is 1.7 MWth. Covering 90% of draws would need 8.3 MWth of backup heat. System LCOE is 56, 71 and 79 €/MWh at the low, median and high cases.

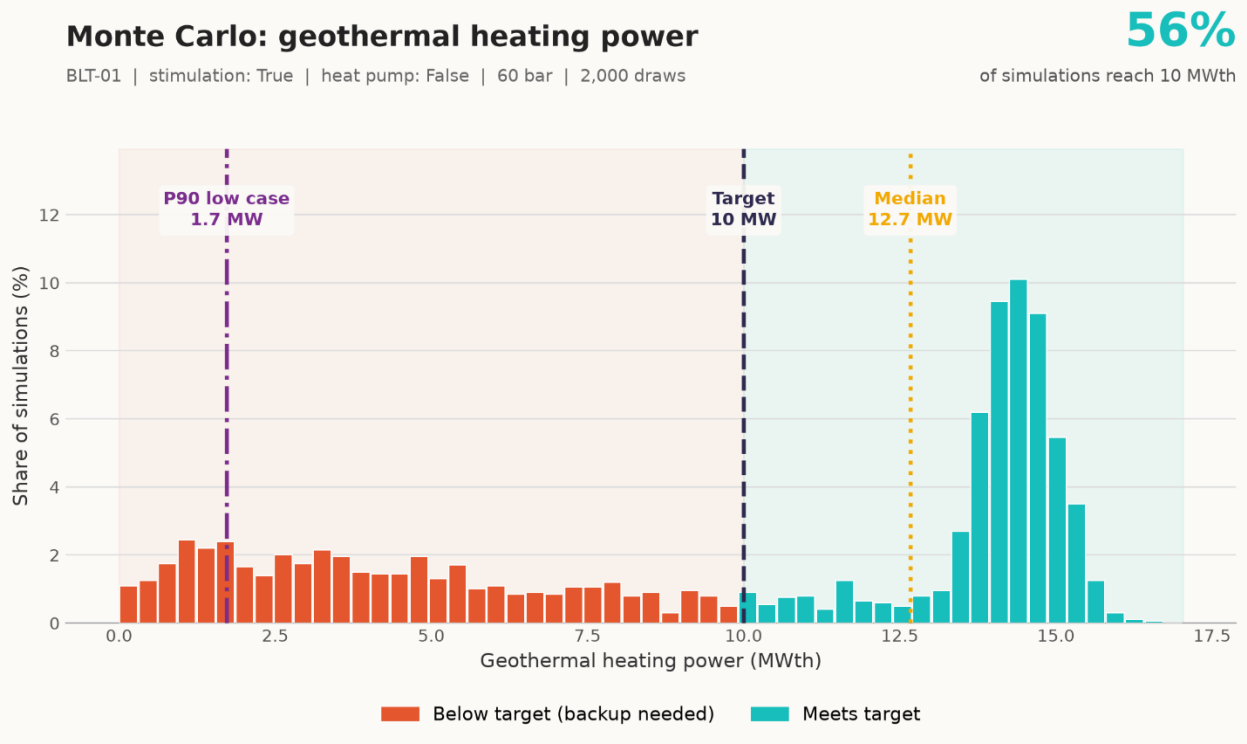


Figure 4 Monte Carlo Simulation on chosen scenario

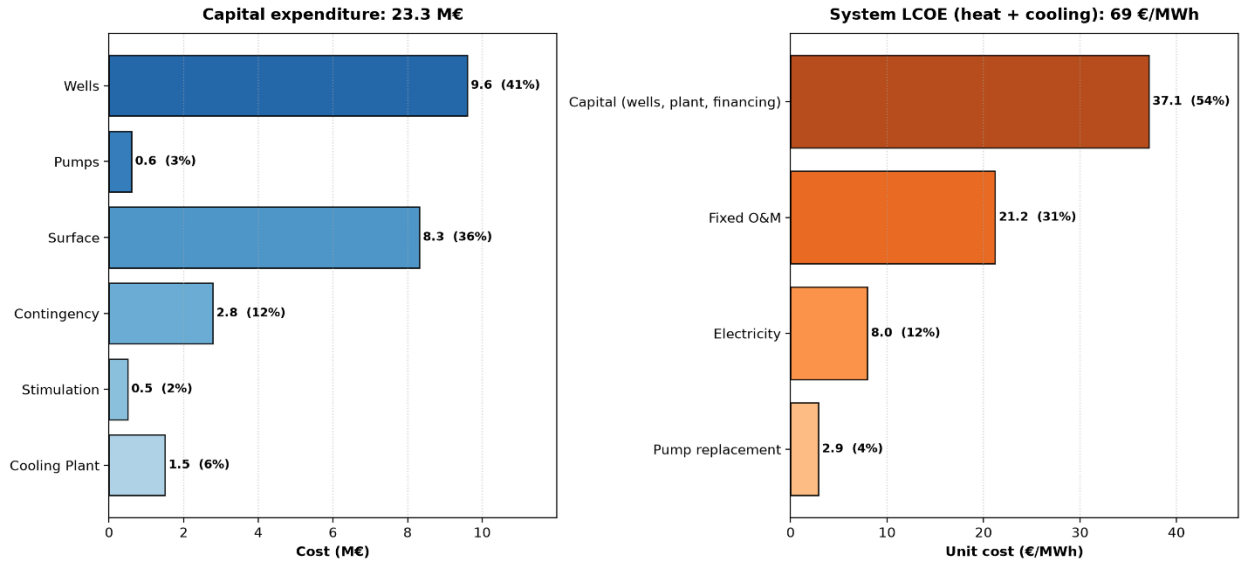


Figure 5 CAPEX and LCOE breakdown

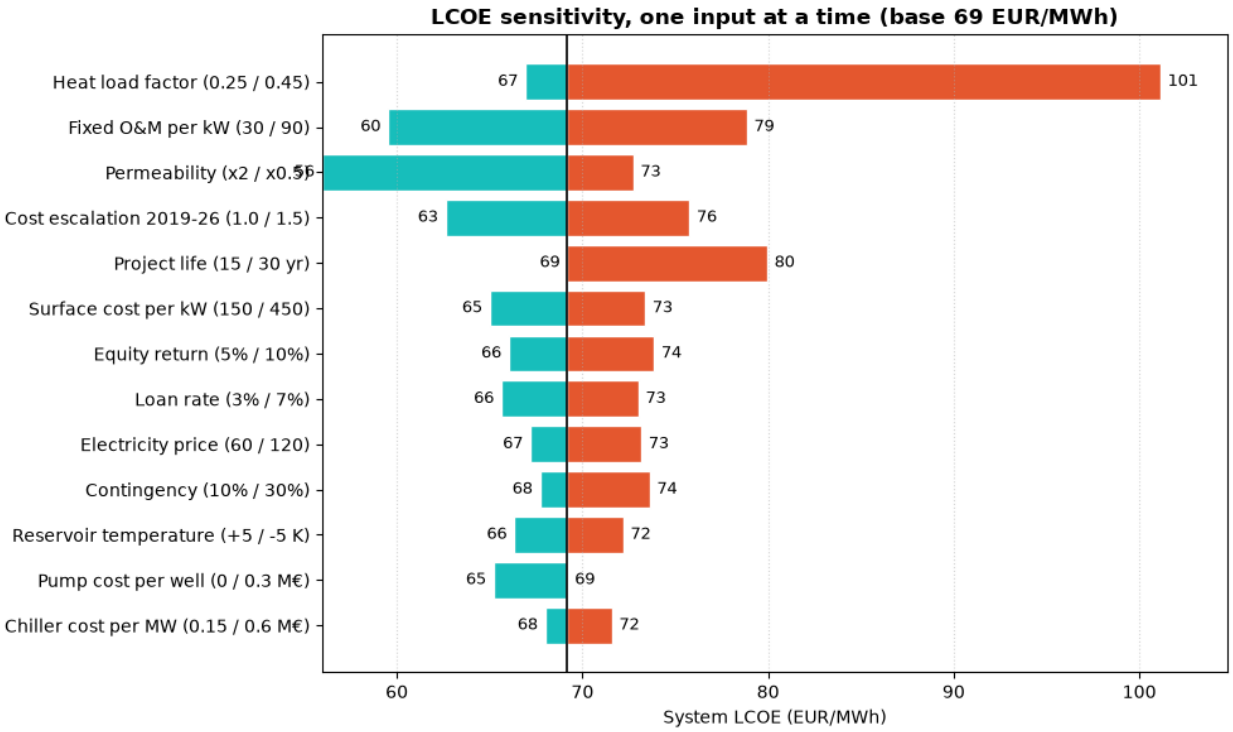


Figure 6 LCOE sensitivity

6. Discussion

6.1 Can the aquifer meet the targets?

For heating, only BLT-01 can carry the target on its own, and only with stimulation. Without intervention it delivers 6.25 MWth, 3.75 MWth short, and JUT-01 delivers 2.21 MWth. With stimulation and a 60 bar drawdown, BLT-01 reaches 12.2 MWth in the deterministic case. Under transmissivity uncertainty, however, the wells alone reach 10 MWth in 56% of draws, and the low case is 1.7 MWth. The target is plausible, not secure.

EVD-01 and PKP-01 can be set aside. Their permeability (6 and 1 mD) and reservoir quality scores are far below BLT-01 and JUT-01, and every sensitivity test kept them at the bottom.

6.2 Which enhancement is worth it?

The two analyses answer different questions. The strategy grid (Table 4) ranks combinations by probability of success using a simple power model with no costs. It favours stimulation plus a heat pump on both wells, at 87%. The system analysis (Table 5) adds capital cost and ranks the scenarios that pass a 50% threshold by capex. It selects a single stimulated doublet because every added heat pump or second doublet raises capital cost by 2 to 24 M€ and the levelised cost to 86 to 152 €/MWh.

The selected design buys the lowest cost at the lowest acceptable reliability. A second doublet with stimulation (Table 5, 60 bar) raises the probability to 75% for 17 M€ more. Whether that is worth paying depends on how costly a shortfall is, which depends on the backup heat source and the contracts that apply. This is the key decision a developer would need to take.

6.3 What about cooling?

The 5 MWth cooling target is met by electric chillers, not by the aquifer. This costs 1.5 M€ and uses electricity at a coefficient of performance of 4.5. It sidesteps the question of whether the aquifer can supply cooling and treats the datathon target as a demand to be served rather than a reservoir output. At base case, BLT-01's 6.25 MWth would exceed 5 MWth if used for heat-driven cooling, but that route was not modelled. An aquifer thermal energy storage alternative is defined in the code and could be a comparison.

6.4 Value of the well-based approach

Combining logs, ThermoGIS values and multi-criteria scoring gives a transparent ranking that holds under every weighting and gradient test. The test is weaker than it looks, because with four wells and one clearly poor well, many reasonable methods will agree. The porosity model is the least secure element: it generalises acceptably to BLT-01 but not to PKP-01, and two of the four wells depend on it.

6.5 Implications for development

The result is a prioritisation tool. The next step is to test BLT-01 directly, since transmissivity is the largest uncertainty and its ThermoGIS range spans a factor of about 50 between P90 and P10. That means a well test, and review of the legacy data and seismic around the well, before any investment decision. Proximity to heat demand, network integration and permitting were outside this analysis.

7. Limitations

- **Few wells.** Four wells give two candidate sites and no area-wide map. ThermoGIS property grids were exported but the ranking used well values, so the conclusions apply near the wells.
- **Porosity.** Only two wells have measured porosity, and the model has R^2 of −0.08 on PKP-01. The 2.02 point uncertainty is reported but porosity does not enter the power calculation, which uses ThermoGIS permeability and transmissivity. It affects only the quality ranking, where it carries the largest weight.
- **Temperature.** The Harrison bottom-hole temperature correction was applied to a gradient-based temperature rather than to a measured bottom-hole temperature, which can overstate reservoir temperature. [State the resulting reservoir temperature at BLT-01 and its sensitivity.]
- **Depth recovery.** Sample depths were recovered by matching exact gamma ray values, with repeated values resolved by keeping the first match.
- **Drawdown.** The selected design uses 60 bar against 18 bar in the base case. That needs justification against injection pressure limits and pumping energy.
- **Uncertainty treatment.** The 32-scenario screening used a log-standard deviation of 0.5, while the final Monte Carlo used 1.53 from BLT-01. The two probabilities (53% and 56%) are therefore not directly comparable.
- **Model simplicity.** The reservoir is homogeneous and steady, with no faults, compartmentalisation or pump failure, and thermal checks rest on a single 150 m3/h case.
- **Economics.** Cost parameters (Table 7) are single point values and none were varied, including the 1.25 escalation to 2026 prices and 80 €/MWh electricity. The levelised cost is a flat nominal price, while the payback calculation indexes the selling price to inflation by default in the code [confirm what the notebook used]. Drilling cost uses one sample depth per well rather than the formation base.
- **Ranking tests.** With four wells, quartile classes place one well in each class by construction, and rank stability is expected.

8. Conclusions and Recommendations

The Slochteren aquifer beneath Utrecht can plausibly deliver the 10 MWth heating target at one location, but not reliably and not cheaply.

1. Only the ROSL_ROSLU unit is present beneath Utrecht, and BLT-01 and JUT-01 are the two viable sites, ranked first and second under every test.
2. At BLT-01 a single stimulated doublet at 60 bar delivers 12.2 MWth and 211 m3/h, with a 56% chance of reaching 10 MWth and a low case of 1.7 MWth.
3. The selected system costs 23.3 M€ and has a levelised cost of 69 €/MWh against 68 €/MWh for the gas alternative, so it is economically marginal.
4. Cooling is met by surface chillers, not by the aquifer.

We recommend a well test at BLT-01 to narrow its transmissivity range, justification of the 60 bar drawdown, and a second doublet or backup heat only if the cost of a shortfall warrants it. Future work could add measured porosity, a heat demand map, area-wide mapping from the ThermoGIS grids, and a comparison with aquifer thermal energy storage for cooling.

Acknowledgements

The author thanks Dr. Daniel Wamriew and Elias Drescher for technical recommendations.

References

1. Kramers, L., van Wees, J.-D., Pluymaekers, M.P.D., Kronimus, A., and Boxem, T. (2012). Direct heat resource assessment and subsurface information systems for geothermal aquifers; the Dutch perspective. *Netherlands Journal of Geosciences*, 91(4).
2. Vrijlandt, M.A.W., Struijk, E.L.M., Brunner, L.G., Veldkamp, J.G., Witmans, N., Maljers, D., and van Wees, J.-D. (2019). ThermoGIS update: a renewed view on geothermal potential in the Netherlands. *European Geothermal Congress 2019*.
3. Harrison, W.E., Luza, K.V., Prater, M.L., and Cheung, P.K. (1983). Geothermal resource assessment of Oklahoma. *Oklahoma Geological Survey Special Publication 83-1*.

4. Wyllie, M.R.J., Gregory, A.R., and Gardner, L.W. (1956). Elastic wave velocities in heterogeneous and porous media. *Geophysics*, 21(1), 41-70.

5. Theis, C.V. (1935). The relation between the lowering of the piezometric surface and the rate and duration of discharge of a well using ground-water storage. *Eos, Transactions AGU*, 16(2), 519-524.

6. Hwang, C.-L., and Yoon, K. (1981). *Multiple Attribute Decision Making: Methods and Applications*. Springer.

7. Diakoulaki, D., Mavrotas, G., and Papayannakis, L. (1995). Determining objective weights in multiple criteria problems: the CRITIC method. *Computers & Operations Research*, 22(7), 763-770.